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LAB 4

1. What is a directive in Assembly Language, and why is it used?

A directive is an instruction given to the assembler to define the program structure and guide how the program should be assembled. It does not generate machine instructions.

2. What is the purpose of the. model directive? Name any three model types.

The. model directive defines the memory model of a program. It tells the assembler how memory is organized.

Three model types are:

1. Small
2. Medium
3. Large

3. What is the purpose of the. stack and .data directives?

- **. stack**: Reserves memory for the stack, which stores temporary data, return addresses, and other values.
- **.data:** Defines the data section where variables, arrays, and constants are declared.

4. Differentiate between the .data and. code directives.

.data	. code
Defines the data section.	Defines the code section.
Used to declare variables and arrays.	Used to write program instructions.
Stores data used by the program.	Contains instructions executed by the CPU.

5. Write any four syntax rules of Assembly Language.

1. Instructions usually contain an operation code (opcode) and operands.
2. Labels are used to identify locations in a program.
3. Comments begin with a semicolon (;).
4. Each instruction should follow the correct Assembly Language syntax.